

Cliff Cezar

Software Developer | AI, Machine Learning, and Automation

Quiapo, Manila, Philippines | (+63) 970 839 8415 | clffczr@gmail.com
github.com/cgcezar | linkedin.com/in/cliff-cezar

Summary

BS Computer Science graduate from the University of the Philippines Baguio with research experience in simulation modeling and data analysis, plus production experience shipping automation features and LLM-powered tooling. Designed and ran a 300-run agent-based simulation study, built the pipeline that processed its output, and used statistical hypothesis testing in R with Python visualization to identify the best-performing intervention. Also integrated an LLM API into a deployed web app to turn unstructured job postings into structured records, and automated scheduling and document generation workflows for a business incubation platform.

Education

Bachelor of Science in Computer Science, University of the Philippines Baguio
Department of Mathematics and Computer Science, Baguio City, Benguet | 2020 – 2026

Research

Applying an Agent-Based Model of COVID-19 Dynamics and Food Relief Systems in Two Barangays in Quiapo, Manila, Undergraduate Thesis, BS Computer Science

University of the Philippines Baguio | Sep 2025 – May 2026

- Designed and built an agent-based simulation model in NetLogo to study COVID-19 transmission in dense urban communities, encoding resident behavior, movement, and contact patterns across two barangays in Quiapo, Manila.
- Ran a 300-run experiment design, 100 iterations for each of three food relief strategies, and built the pipeline that collected, cleaned, and aggregated the resulting simulation datasets for analysis.
- Applied statistical hypothesis testing in R to compare outcomes across strategies and identify which relief scheme most reduced disease spread, then produced the result figures in Python using pandas and matplotlib.
- Documented the full methodology, parameter settings, and findings so the experiment is reproducible, and defended the completed study.

Experience

Software Developer, SILBI Web App Project

Baguio City, Benguet | Feb 2025 – May 2025

- Built and deployed a centralized platform for business incubatees on Next.js, TypeScript, and PostgreSQL with Prisma, covering the full lifecycle from requirements gathering to production deployment.
- Automated roadmap tracking, meeting scheduling, and legal document generation, replacing repetitive manual admin work with backend workflows.
- Designed the relational schema and wrote the queries behind records, scheduling, and an activity audit log, and implemented role-based access control for user, mentor, and admin accounts.
- Containerized the application with Docker Compose behind an nginx reverse proxy with TLS, with separate development and production configurations on Linux.
- Live site: upbsilbitulay.vercel.app | Source: github.com/rye0x/silbi-tulay

Projects

Job-Tag: LLM-Assisted Job Application Tracker, Developer

Personal Project | 2026

- Integrated an LLM API through a serverless function that reads a pasted job posting link and returns structured fields for job title, company, work setup, and employment type, removing most of the manual encoding per application.
- Engineered the prompt and response handling to return consistent structured output, with validation and fallbacks so malformed or incomplete model responses do not break the app.
- Swapped model providers without changing application code by isolating the inference call behind a single serverless endpoint.
- Built the surrounding full-stack app in React and Node with a scheduling calendar and document upload, and set up a GitHub Actions CI/CD pipeline for automated build and deployment.
- Live site: job-tag.vercel.app | Source: github.com/cgcezar/job-tag

Byte: A Coding Pet, Developer

Personal Project | 2026

- Built a browser-based virtual pet game in React and Vite where every interaction is earned by solving a JavaScript coding challenge instead of clicking a button.
- Implemented an automated evaluation harness that executes user-submitted functions against test cases in a contained scope, scoring each submission per test case and handling syntax and runtime errors without breaking the app.
- Designed an XP and evolution system with three difficulty tiers and decaying pet stats, keeping game logic as plain framework-independent JavaScript separate from React state and UI components.
- Source: github.com/cgcezar/byte-coding-pet

Additional shipped sites: UP Baguio Mountaineers website and Open Climb 2026 event site, built in plain HTML, CSS, and JavaScript with Leaflet and FullCalendar (cgcezar.github.io/upbm-website); Rotaract Club of UP Baguio organization website (cgcezar.github.io/rotaract-upb-website)

Technical Skills

Machine Learning & Data: agent-based modeling and simulation (NetLogo), experiment design and parameter sweeps, statistical hypothesis testing in R, data cleaning and aggregation, pandas, matplotlib, data visualization

AI & LLM: LLM API integration, prompt design for structured output, serverless inference endpoints, sentence-transformer embeddings, retrieval-augmented generation concepts

Automation & DevOps: GitHub Actions CI/CD, Docker and Docker Compose, nginx, serverless functions, AWS, Linux and UNIX shell, Git and GitHub

Programming: Python, R, JavaScript, TypeScript, SQL, Java, PHP, C/C++, MATLAB

Web & Backend: Next.js, React, Node.js, REST APIs, PostgreSQL, Prisma, HTML, CSS, Vercel

Tools: LaTeX, Figma, QGIS, ArcGIS, Canva

Languages

Onhan (Native), Filipino (Fluent), English (Fluent), Japanese (Basic)

Interests

Mountaineering, Cartography, Cooking, Photography, Ultimate Frisbee